

GENETIC ALGORITHMS

[5]

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# Simple Genetic Algorithm
# Goal: Find a number between 0 and 10
# that gives the highest x2 value

import random

# Fitness function
def fitness(x):
    return x ** 2

# Step 1: Create an initial population
population = [random.randint(0, 10) for _ in range(6)]

print("Initial Population:", population)

# Repeat the process for 5 generations
for generation in range(5):

    print("\nGeneration:", generation + 1)

    # Step 2: Arrange numbers based on fitness
    # Higher fitness comes first
    population.sort(key=fitness, reverse=True)

    print("Population:", population)
    print("Fitness values:", [fitness(x) for x in population])

    # Step 3: Select the two best parents
    parent1 = population[0]
    parent2 = population[1]

    print("Selected Parents:", parent1, parent2)

    # Keep the best parents
```

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print("Selected Parents:", parent1, parent2)

# Keep the best parents
new_population = [parent1, parent2]

# Step 4: Create children
while len(new_population) < 6:

    # Crossover:
    # Take the average of the parents
    child = (parent1 + parent2) // 2

    # Step 5: Mutation:
    # Randomly increase or decrease the child
    mutation = random.choice([-1, 0, 1])
    child = child + mutation

    # Keep child between 0 and 10
    child = max(0, min(10, child))

    new_population.append(child)

# Step 6: Replace old population
population = new_population

# Find the final best solution
population.sort(key=fitness, reverse=True)

best_value = population[0]

print("\nFinal Best Value:", best_value)
print("Final Best Fitness:", fitness(best_value))
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Output:

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. Initial Population: [7, 1, 6, 6, 8, 3]

Generation: 1
Population: [8, 7, 6, 6, 3, 1]
Fitness values: [64, 49, 36, 36, 9, 1]
Selected Parents: 8 7

Generation: 2
Population: [8, 8, 7, 7, 6, 6]
Fitness values: [64, 64, 49, 49, 36, 36]
Selected Parents: 8 8

Generation: 3
Population: [8, 8, 8, 8, 7, 7]
Fitness values: [64, 64, 64, 64, 49, 49]
Selected Parents: 8 8

Generation: 4
Population: [9, 8, 8, 7, 7, 7]
Fitness values: [81, 64, 64, 49, 49, 49]
Selected Parents: 9 8

Generation: 5
Population: [9, 9, 8, 7, 7, 7]
Fitness values: [81, 81, 64, 49, 49, 49]
Selected Parents: 9 9

Final Best Value: 10
Final Best Fitness: 100
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